

Bayesian Enhanced Baseline-and-Residual Model for Advanced Basketball Analytics

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Abstract

We propose a novel "baseline-and-residual" model for evaluating basketball players' on-court impact, called the Bayesian Enhanced Baseline-and-Residual Model (BEBRM). This framework integrates a rich set of non-traditional performance metrics (defensive matchup results, hustle stats, spatial spacing impact, Synergy play-type data, etc.) into a Bayesian prior baseline, and then applies a multi-round residual correction (reminiscent of a regularized plus-minus adjustment) at the possession level. The Bayesian baseline shrinks extreme estimates for players with limited data, providing more reliable initial ratings and credible intervals. The iterative residual adjustment then distributes each possession's outcome across the players on the court, refining individual impact estimates based on in-game context. The result is a holistic player impact metric that aims to reduce reliance on raw box-score "inflation" and highlight under-the-radar contributions (e.g. screen assists, forced turnovers, hockey assists) that typical metrics may miss.

1 Introduction

Advanced basketball analytics has long grappled with the challenge of separating genuine player impact from superficial box-score accumulation. Since the introduction of adjusted plus-minus by Rosenbaum [1] and the development of early statistical models by Kubatko et al. [2], numerous approaches have been proposed to better measure player contributions. Regularized Adjusted Plus/Minus (RAPM) introduced ℓ_2 -regularization to stabilize on/off ratings [3], and a Bayesian hierarchical variant further improved estimates via partial pooling [4]. Box Plus/Minus (BPM) shifted focus back to box-score data as predictors of impact [5]. More recently, hybrid metrics such as Estimated Plus/Minus (EPM) [6] and LEBRON [7] combine box-score-based priors with on/off lineup data. Each of these methods provides only a partial solution: some rely heavily on box-score statistics, while others depend purely on on/off differentials. Small sample sizes (e.g. rookies or players with few minutes) and outlier performances remain problematic, often leading to noisy or misestimated impact ratings.

In this paper, we propose a Bayesian Enhanced Baseline-and-Residual Model (**BEBRM**) to holistically estimate players' contributions to team success by integrating rich non-box-score information with possession-level adjustments. In the first stage, a Bayesian baseline model incorporates a broad set of features—including defensive matchup outcomes, hustle events, spacing/spatial metrics, and Synergy-derived play-type efficiencies—to establish an informed prior estimate of each player's offensive and defensive impact. This prior uses hierarchical modeling to account for group differences (e.g. by player position) and applies a Bayesian ridge regression to handle multicollinearity, shrinking extreme estimates toward reasonable values. In the second stage, the model applies

an iterative regularized plus-minus adjustment at the possession level, distributing each possession’s residual (the deviation of the actual outcome from the expected baseline) across the players on the court. Successive ridge regressions on possession outcomes refine each player’s rating based on context-dependent performance (e.g. lineup synergy and opponent strength). The end result is a player impact estimate that blends long-term indicators from the prior with in-game situational adjustments, yielding a more robust measure of a player’s overall influence on team offensive and defensive efficiency.

2 Data Sources

Our analysis draws on multiple data sources from the NBA’s tracking and statistics services to construct each player’s feature set. Let N be the set of players evaluated (for example, all players in a given season). The key data sources are summarized below:

- **Play-by-Play Data:** Possession-level logs of every game (made/missed shots, turnovers, substitutions, etc.) – used to identify which players are on/off the floor each possession and the outcome (points scored).
- **Matchup Data:** Defensive matchup statistics (e.g. opponent field goal attempts and makes when guarded, help defense events, forced turnovers in matchups). These provide granular insight into a player’s direct defensive impact.
- **Hustle Stats:** Hustle play data (contested shots, deflections, charges drawn, screen assists, etc.), capturing intangible contributions that don’t appear in a standard box score.
- **Shot Tracking:** Spatial shot chart data (locations of shots, shot distances, outcomes) for spacing metrics and ”gravity” – e.g. how a player’s presence affects the distribution of team-mate shots (more open threes, rim attempts, etc.).
- **On/Off and Advanced Box Score:** Team efficiency splits (offensive/defensive ratings when a player is on vs. off the court), Four Factors (rebound%, turnover%, etc.), and other advanced stats (true shooting%, usage rate). These provide high-level contextual indicators of impact.
- **Synergy Play-Type Data:** Performance in specific play types (isolation, pick-and-roll ball-handler, spot-up shooting, post-up defense, etc.), including points-per-possession and frequency for each category. This offers specialized insight into a player’s offensive and defensive role effectiveness.

Each player’s feature vector is built by merging these data sources. In practice, for each player $i \in N$ we construct an expanded feature vector:

$$\mathbf{X}_i = [X_{i,1}, X_{i,2}, \dots, X_{i,m}] ,$$

where each component $X_{i,k}$ corresponds to a metric derived from the above endpoints (matchup stats, hustle stats, synergy metrics, etc.). For example, features can include a player’s isolation efficiency (points per possession), pick-and-roll efficiency, screen assists per minute, opponent field goal percentage when this player is the nearest defender, on/off net rating impact, usage rate, and so on. This rich feature set forms the input to the baseline model.

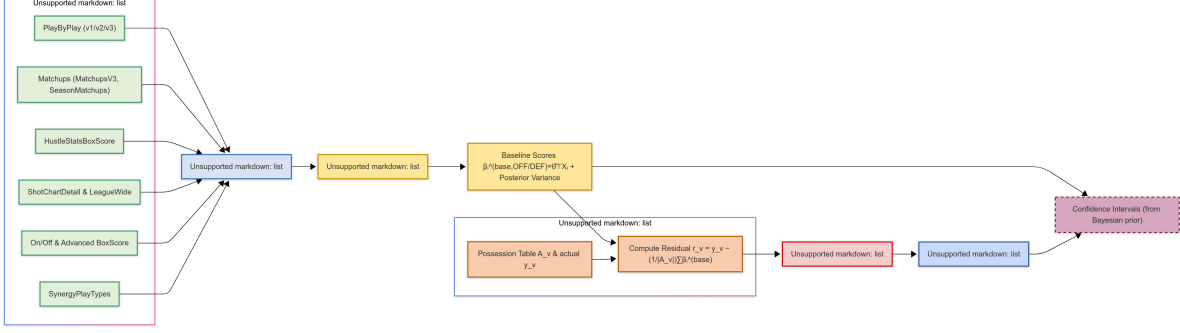


Figure 1: *Schematic of the BEBRM modeling process.* Various data sources (green boxes on the left) feed into the feature construction for each player, which the Bayesian baseline model uses to produce initial offensive/defensive ratings (β_i^{base}) along with uncertainty estimates. Possession-level data (blue box) is then used to compute residuals r_v for each possession, which are iteratively allocated to players via ridge regression updates (red boxes). The model outputs final player ratings β_i^{final} and confidence intervals (dotted outline) from the Bayesian prior step.

3 Bayesian Baseline Prior

3.1 Bayesian Ridge Regression on Multi-Source Features

In the baseline stage, we fit a Bayesian regularized regression to predict an initial performance outcome for each player from their feature vector. Let Y_i be a chosen response metric for player i that represents overall impact (for example, one could use a prior estimate of the player’s plus-minus or a composite impact score derived from team performance). We assume a linear model:

$$Y_i = \mathbf{X}_i^\top \boldsymbol{\theta} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2), \quad (1)$$

where $\boldsymbol{\theta}$ is the coefficient vector linking features to the response. We place a Gaussian prior on $\boldsymbol{\theta}$:

$$\boldsymbol{\theta} \sim N(\mathbf{0}, \alpha^2 I), \quad (2)$$

which serves as a Bayesian ridge regularization (with prior variance α^2 controlling the degree of shrinkage). This Bayesian formulation ensures that the regression coefficients (and thus the contributions of each feature) are shrunk toward zero, preventing overfitting especially for features based on limited data.

Given observations $\{(\mathbf{X}_i, Y_i)\}_{i=1}^N$, the posterior distribution for $\boldsymbol{\theta}$ is proportional to the likelihood times the prior:

$$p(\boldsymbol{\theta} \mid \{\mathbf{X}_i, Y_i\}_{i=1}^N) \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^N (Y_i - \mathbf{X}_i^\top \boldsymbol{\theta})^2 - \frac{1}{2\alpha^2} \|\boldsymbol{\theta}\|^2 \right\}, \quad (3)$$

which is also the objective for ridge regression. The posterior mean $\hat{\boldsymbol{\theta}}$ can be obtained in closed form (as in standard ridge regression) or via Bayesian sampling methods. Intuitively, $\hat{\boldsymbol{\theta}}$ provides the best-fitting linear combination of the multi-source features for predicting Y , while heavily shrinking the contributions of noisy or collinear features.

We fit this model separately for offense and defense to obtain two sets of coefficients. Let $\hat{\boldsymbol{\theta}}^{(\text{off})}$ be the learned coefficients from an offensive regression (predicting an offensive impact measure

$Y_i^{(\text{off})}$), and $\hat{\boldsymbol{\theta}}^{(\text{def})}$ from a defensive regression (predicting a defensive impact measure $Y_i^{(\text{def})}$). We can then compute each player’s baseline offensive and defensive ratings by plugging their feature vector into these models. Specifically, for each player i :

$$\beta_i^{(\text{base, OFF})} = \hat{\boldsymbol{\theta}}^{(\text{off})\top} \mathbf{X}_i^{(\text{off})}, \quad \beta_i^{(\text{base, DEF})} = \hat{\boldsymbol{\theta}}^{(\text{def})\top} \mathbf{X}_i^{(\text{def})}, \quad (4)$$

where $\mathbf{X}_i^{(\text{off})}$ and $\mathbf{X}_i^{(\text{def})}$ are the subsets of features relevant to offense and defense respectively. These β_i^{base} values represent our model’s prior estimates of player i ’s offensive and defensive impact (e.g. in terms of points contributed or saved per 100 possessions, depending on how Y is defined).

Crucially, the Bayesian framework provides a measure of uncertainty for these baseline ratings. The posterior covariance of $\boldsymbol{\theta}$ from Eq. (3) can be used to derive a credible interval for each player’s β_i^{base} . In practice, players with very limited data (for example, a rookie who has played few minutes) will have features that contribute with large uncertainty, and their baseline rating β_i^{base} will be heavily shrunk towards an average value (near 0 impact) with a wide credible interval. In contrast, a veteran player with abundant data will have a more confident baseline estimate. This addresses small-sample volatility by automatically regressing uncertain cases toward the mean and explicitly quantifying our confidence in each player’s baseline impact.

4 Possession-Level Residual Correction

4.1 Motivation

While the baseline ratings $\beta_i^{(\text{base, OFF/DEF})}$ incorporate a wealth of information about a player (synergy stats, hustle plays, etc.), they do not explicitly account for the context of which teammates or opponents share the floor with that player on each possession. In other words, the baseline provides an expected impact for a player in general, but game-by-game performance will also depend on lineup interactions and opponent matchups. To capture these context-dependent effects, we implement a possession-level adjustment reminiscent of plus-minus analysis: we look at each possession’s outcome relative to what the baseline expectations would predict, and allocate the residual difference to the players involved in that possession.

4.2 Model Setup for Offensive Possessions

Let N_{pos} be the total number of offensive possessions in the dataset (across all games and teams considered). For each offensive possession ν (where $1 \leq \nu \leq N_{\text{pos}}$), define:

- A_ν = the set of offensive players on the court during possession ν (typically 5 players for the team on offense).
- $y_\nu^{(\text{OFF})}$ = points scored on that possession by the offensive team (this can be the actual points scored; one may also use a "luck-adjusted" value that, for instance, adjusts 3-point shots toward their expected value to reduce noise, though we will treat it as actual points here).

First, we compute the baseline expected points for possession ν as the average of the baseline offensive ratings of the players on the floor:

$$\hat{y}_\nu^{(0, \text{OFF})} = \frac{1}{|A_\nu|} \sum_{i \in A_\nu} \beta_i^{(\text{base, OFF})}, \quad (5)$$

where $\beta_i^{(\text{base}, \text{OFF})}$ comes from the baseline model (Eq. (4)). This $\hat{y}_\nu^{(0, \text{OFF})}$ is essentially what the model expects the offense to score on that possession, given the offensive talent on the floor, before considering any possession-specific context. The *residual* for possession ν is then defined as:

$$r_\nu^{(\text{OFF})} = y_\nu^{(\text{OFF})} - \hat{y}_\nu^{(0, \text{OFF})} , \quad (6)$$

i.e. the difference between actual points scored and the baseline expected points.

4.3 Iterative Residual Allocation via Ridge Regression

We now allocate these possession-level residuals $r_\nu^{(\text{OFF})}$ back to the players on the court, as adjustments to their ratings. We employ an iterative ridge regression procedure to ensure the allocation is stable and regularized:

1. **Initialization:** For all players i , set $\beta_i^{(0, \text{OFF})} = \beta_i^{(\text{base}, \text{OFF})}$. In other words, start the iterative process with each player's offensive rating equal to their baseline rating from the first stage.
2. **Iterative update:** At iteration t (starting from $t = 1$), compute the residual for each possession ν with the current ratings:

$$r_\nu^{(t, \text{OFF})} = y_\nu^{(\text{OFF})} - \frac{1}{|A_\nu|} \sum_{i \in A_\nu} \beta_i^{(t-1, \text{OFF})} ,$$

which is analogous to Eq. (5) but using the latest ratings. Then, we set up a ridge regression to explain these residuals by assigning an additive adjustment Δ_i to each player i . Let $x_{\nu, i} = 1$ if player i was on offense in possession ν (i.e. $i \in A_\nu$), and $x_{\nu, i} = 0$ otherwise. We solve for the adjustments $\{\Delta_i\}$ by minimizing:

$$\sum_{\nu=1}^{N_{\text{pos}}} \left(r_\nu^{(t, \text{OFF})} - \sum_i x_{\nu, i} \Delta_i \right)^2 + \lambda_{\text{OFF}} \sum_i \Delta_i^2 , \quad (7)$$

where λ_{OFF} is a ridge regularization parameter for the residual allocation. This is a linear ridge regression problem across all possessions: we are finding player-specific adjustments Δ_i that best account for the residual outcomes, with regularization to avoid overfitting a single possession or small sample effects. The solution can be obtained in closed form or with standard ridge regression solvers.

3. **Rating update:** For each player i , update their offensive rating estimate as

$$\beta_i^{(t, \text{OFF})} = \beta_i^{(t-1, \text{OFF})} + \rho \Delta_i , \quad (8)$$

where ρ is a step size (we can set $\rho = 1$ for a full update in each iteration, or a smaller ρ if we want to take more gradual steps). Here Δ_i is the solution from the ridge regression in Eq. (7).

4. **Repeat:** Iterate steps (a)–(c) until convergence or until a fixed number of iterations T is reached. In practice, this process typically stabilizes in a few iterations; the regularization λ_{OFF} ensures that the adjustments Δ_i are modest for each round.

After the final iteration T , we obtain the refined offensive ratings $\beta_i^{(T, \text{OFF})}$ for each player. We denote these as the final offensive impact ratings:

$$\beta_i^{(\text{final}, \text{OFF})} = \beta_i^{(T, \text{OFF})} .$$

This multi-round approach means that we did not rely on a single pass of the residuals; instead, the model gradually adjusts and "shares" credit among players over successive passes. Notably, if a player had very limited data and thus a baseline $\beta_i^{(\text{base}, \text{OFF})}$ near zero, they will not immediately jump to a high rating from one or two good possessions. Only consistent residual outperformance over many possessions will significantly raise that player's rating. This guards against overreacting to small sample performance spikes.

We apply an analogous procedure for defensive possessions. For each defensive possession (where the player is on defense), define B_ν as the set of defensive players on the court and $y_\nu^{(\text{DEF})}$ as the points allowed by the defense on possession ν . We compute residuals

$$r_\nu^{(t, \text{DEF})} = y_\nu^{(\text{DEF})} - \frac{1}{|B_\nu|} \sum_{j \in B_\nu} \beta_j^{(t-1, \text{DEF})} ,$$

and similarly solve a ridge regression to distribute these defensive residuals among the players $j \in B_\nu$, yielding adjustments Δ_j to update $\beta_j^{(t, \text{DEF})}$. This yields final defensive ratings $\beta_i^{(\text{final}, \text{DEF})}$ for each player.

4.4 Final Ratings Output

After the offensive and defensive iterative adjustments have converged, we have for each player i a final offensive rating $\beta_i^{(\text{final}, \text{OFF})}$ and defensive rating $\beta_i^{(\text{final}, \text{DEF})}$. We can report these separately (to highlight offensive vs. defensive contributions), or sum them to get an overall impact rating:

$$\beta_i^{(\text{final}, \text{TOTAL})} = \beta_i^{(\text{final}, \text{OFF})} + \beta_i^{(\text{final}, \text{DEF})} , \tag{9}$$

which would represent an estimate of the player's total net impact on point differential (offense minus defense). Along with these point-estimates, we carry forward the uncertainty from the baseline stage: the credible interval from the baseline can be viewed as a measure of confidence in the rating, where a player with a short track record will have a wider interval, signaling that the estimate should be taken with caution.

5 Discussion: Design Advantages and Limitations

5.1 Small-Sample Stability

One of the key design advantages of the BEBRM approach is its expected stability in small-sample situations. By incorporating a Bayesian prior, the baseline stage inherently pulls extreme values toward the mean for players with scant data. This should prevent a little-used player from being unjustifiably rated as elite or terrible based on a few events. The iterative residual correction further ensures that no single game or single possession can overly swing a player's rating; adjustments are spread out over many observations.

5.2 Capturing Non-Box-Score Contributions

Another intended strength of BEBRM is its ability to highlight contributions that do not show up in the traditional box score. Because the baseline feature vector $\mathbf{X}_i$ includes Synergy play-type efficiencies, hustle stats like screen assists and deflections, and spacing metrics from shot charts, a player who excels in these areas should receive credit in the baseline rating even if their points, rebounds, or assists are modest. For example, a forward who sets excellent screens and makes the extra pass might boost teammates’ scoring without scoring much themselves; our baseline model would likely give this player a solid offensive prior due to high screen assist rates and on/off offensive rating boosts. The residual adjustment stage then adds the context of actual on-court results to fine-tune this evaluation. In summary, BEBRM’s design places less emphasis on raw box-score totals and more on context-driven or intangible strengths, which should help surface players who are impactful in subtle ways.

5.3 Uncertainty and Credible Intervals

Unlike many traditional metrics, BEBRM provides a measure of uncertainty (through the Bayesian posterior). This is valuable in practice: an analyst or team decision-maker can understand not just a player’s point estimate rating but also how confident the model is in that rating. For instance, if a little-known player ranks surprisingly high, BEBRM would also show a wide credible interval, signaling caution. In a front-office discussion, this can temper overreaction to a small-sample standout by explicitly showing the estimate’s uncertainty. Conversely, established stars would have very tight credible intervals, reinforcing confidence in their high rating. This probabilistic insight is a distinct advantage of the Bayesian approach used in the baseline stage.

5.4 Structural Comparison to EPM and LEBRON

BEBRM shares a philosophical similarity with metrics like EPM and LEBRON in that all three use some form of prior plus an on/off adjustment. However, there are important differences in structure and data usage.

EPM (Estimated Plus-Minus) typically builds a single-stage model: it uses a set of box-score and player tracking features to predict impact (often via linear or nonlinear regression) and then performs a one-pass lineup adjustment (often Bayesian updating or regression) to get final ratings. EPM’s prior is usually informed by box-score and tracking data, but it may not explicitly incorporate certain play-by-play derived stats like specific matchup outcomes or some of the more granular Synergy categories. It also generally does not iterate the on/off adjustment multiple times – it’s a one-shot blend of the prior and plus-minus. In contrast, BEBRM uses a broader feature set (including those hustle and play-type stats EPM might not include) and an explicit Bayesian ridge prior for shrinkage. We then apply an iterative residual allocation (multiple passes) instead of a single pass.

LEBRON (an acronym for "Luck-adjusted Box-score prior Regularized ON/Off") is another contemporary metric that is structurally similar to EPM. LEBRON starts with a box-score weighted prior metric (often called "BoxLEBRON") and then performs a RAPM-style regression to adjust for on/off data, usually in one step. It is termed "luck-adjusted" because it attempts to adjust for random variance (like opponent 3-point shooting luck) in the on/off calculation. However, LEBRON’s prior, while more sophisticated than raw box score (it includes some advanced weighted combinations of stats), does not incorporate the full range of non-traditional metrics either. Moreover, LEBRON’s on/off adjustment is typically a single pass rather than an iterative one. It does not employ a Bayesian framework for the prior—meaning it doesn’t provide an explicit posterior

distribution for the player ratings or formally shrink the prior beyond whatever regularization is used in the RAPM step.

LEBRON’s approach is effectively a special case of our framework: it is as if LEBRON had a simpler prior and set the number of residual iterations to one. The trade-off is that BEBRM is more data-hungry and complex, but it can potentially identify subtle player impacts (through rich features) that LEBRON might miss, and it provides the extra benefits of iterative refinement and uncertainty quantification.

5.5 Limitations and Future Work

BEBRM’s distinguishing features relative to models like EPM and LEBRON are the breadth of data it leverages and its explicit Bayesian iterative framework. It strives to combine the best aspects of box-score evaluation and plus-minus analysis. However, the cost of this approach is greater complexity and data requirements. The model requires access to multiple NBA data sources (Synergy, hustle stats, matchup data, etc.) that may not always be publicly available. Additionally, the iterative ridge regression procedure adds computational overhead compared to single-pass methods.

Future work should focus on empirical validation using real NBA data to assess whether the theoretical advantages translate into practical improvements. Extensions could include hierarchical structures (to handle team or lineup effects, or rookie vs. veteran priors) and exploration of the model’s predictive power for game outcomes or player development trajectories.

6 Conclusion

We introduced **BEBRM**, a Bayesian Enhanced Baseline-and-Residual Model, as a comprehensive framework for advanced basketball analytics. This model marries a rich feature-based prior with an iterative plus-minus residual adjustment to yield player impact ratings that are both data-informed and context-sensitive. In the baseline stage, a Bayesian regression incorporating defensive matchup data, hustle statistics, play-type efficiencies, and shot-spacing metrics addresses the biases of limited sample sizes by shrinking estimates toward reasonable values and providing confidence intervals. In the second stage, a multi-round ridge regression on possession-level residuals refines each player’s rating based on actual on-court performance in context, allocating credit (or blame) for team outcomes to the players involved in a stable, regularized way.

It is important to note that each player evaluation metric comes with its own assumptions and areas of focus. Traditional metrics like BPM prioritize individual stats, RAPM focuses purely on on/off outcomes, and newer hybrids like LEBRON and EPM attempt to blend these perspectives. BEBRM contributes to this landscape by explicitly incorporating non-traditional data and Bayesian thinking, but it does not render other metrics obsolete. In practice, no single metric is universally superior—each has its rationale and can be useful depending on the context. Front offices and analysts should consider a suite of metrics, including BEBRM, LEBRON, and EPM, to get a well-rounded view of player value. Each model highlights different facets of performance, and areas where they agree are likely robust findings, whereas divergences can prompt deeper investigation.

In closing, BEBRM offers a fresh, holistic framework for player evaluation, integrating “eye-test” stats (hustle, matchups) with advanced modeling. Its design advantage lies in balancing small-sample caution with recognition of intangibles, providing both an estimated impact and a measure of confidence. By acknowledging uncertainty and context, BEBRM aims to move us a step closer to a nuanced understanding of player contributions. Nonetheless, it complements rather than replaces existing metrics, and the wise analyst will use it alongside other tools to make informed decisions.

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